

20 GitHub Repositories for DevOps Practice

A curated practice set covering Java, Python, Node.js, C++, .NET, containers, Kubernetes, CI/CD, GitOps, and observability.

Recommended DevOps progression

Source → Build → Test → Security → Docker → Registry → Kubernetes → Helm → CI/CD → GitOps → Monitoring

#	Repository	Technology	Level	GitHub link
1	Spring PetClinic	Java / Spring	Beginner	https://github.com/spring-projects/spring-petclinic
2	Flask	Python	Beginner	https://github.com/pallets/flask
3	Express.js	Node.js	Beginner	https://github.com/expressjs/express
4	.NET eShop	C# / .NET	Beginner	https://github.com/dotnet/eShop
5	CMake Examples	C++	Beginner	https://github.com/troy50/cmake-examples
6	Spring PetClinic Microservices	Java / Spring Cloud	Intermediate	https://github.com/spring-petclinic/spring-petclinic-microservices
7	Spring PetClinic REST	Java / REST	Intermediate	https://github.com/spring-petclinic/spring-petclinic-rest
8	Django	Python	Intermediate	https://github.com/django/django
9	FastAPI	Python	Intermediate	https://github.com/fastapi/fastapi
10	Node.js	Node.js / C++	Intermediate	https://github.com/nodejs/node
11	NestJS	Node.js / TypeScript	Intermediate	https://github.com/nestjs/nest
12	ASP.NET Core	C# / .NET	Intermediate	https://github.com/dotnet/aspnetcore
13	Spring PetClinic on Istio	Java / Kubernetes / Istio	Advanced	https://github.com/spring-petclinic/spring-petclinic-istio
14	Kubernetes Examples	Kubernetes / Multiple	Advanced	https://github.com/kubernetes/examples
15	Awesome Compose	Docker / Multiple	Advanced	https://github.com/docker/awesome-compose
16	Prometheus	Go	Advanced	https://github.com/prometheus/prometheus
17	Grafana	Go / TypeScript	Advanced	https://github.com/grafana/grafana
18	Argo CD	Go / Kubernetes	Advanced	https://github.com/argoproj/argo-cd
19	Jenkins	Java	Advanced	https://github.com/jenkinsci/jenkins
20	OpenTelemetry Demo	Multi-language / Microservices	Advanced	https://github.com/open-telemetry/opentelemetry-demo

Best repositories to start with

- 1. Spring PetClinic** — strong starting point for Maven/Gradle, Java builds, tests, containers and Kubernetes. The current repository includes Maven/Gradle builds, Kubernetes files and Docker Compose support.
- 2. Spring PetClinic Microservices** — excellent intermediate/advanced project for microservices, service discovery, API gateway, tracing, Prometheus and Grafana.
- 3. OpenTelemetry Demo** — excellent advanced project for distributed systems and observability across multiple languages.

Suggested hands-on exercise

For each application, create your own CI/CD pipeline rather than simply running the existing pipeline. Start with build and unit tests, then add SonarQube/static analysis, dependency scanning, Docker image builds, image vulnerability scanning, a container registry, Kubernetes manifests, Helm, and finally GitOps with Argo CD. Add Prometheus/Grafana or OpenTelemetry for monitoring and tracing.

Note: Repository availability and contents can change over time. The links in this PDF point directly to the public GitHub repositories.